Capstone failure testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Capstone failure testing before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone service implementation

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Red team exercises

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accessibility and user feedback

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Practise with fading help

> Complete the supported Capstone failure testing trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone failure testing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Capstone failure testing explanation against the supplied facts before discussing it.

Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Capstone failure testing facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Capstone failure testing facts and keep your reasoning.

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Capstone failure testing facts and keep your reasoning.

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Capstone failure testing facts and keep your reasoning.

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Capstone failure testing facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Capstone failure testing to Grounded Learning Service, identify a limit and choose the next useful test.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup.

Compute case pass rate and release implication. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Capstone failure testing

> Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval. Explain the labelled step before checking the changed condition.

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

AI ACADEMY / AI-6/34

Capstone failure testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Verify source

Retrieval supplies F4 and the time 15:00.



2. Inject failure

Disable drafting without changing source evidence.



3. Use bounded fallback

Return the checked template citing F4 and record degraded operation.

CHANGE ONE THING

Make retrieval unavailable as well.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the checked template citing F4 and record degraded operation.

> 1. Verify source

2. Use bounded fallback

3. Inject failure

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Use bounded fallback Return the checked template citing F4 and record degraded operation.

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Make retrieval unavailable as well. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Return unsupported or unavailable; do not use the factual template.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

Repair a missing building block

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Red team exercises (ai-6/31). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction.

Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

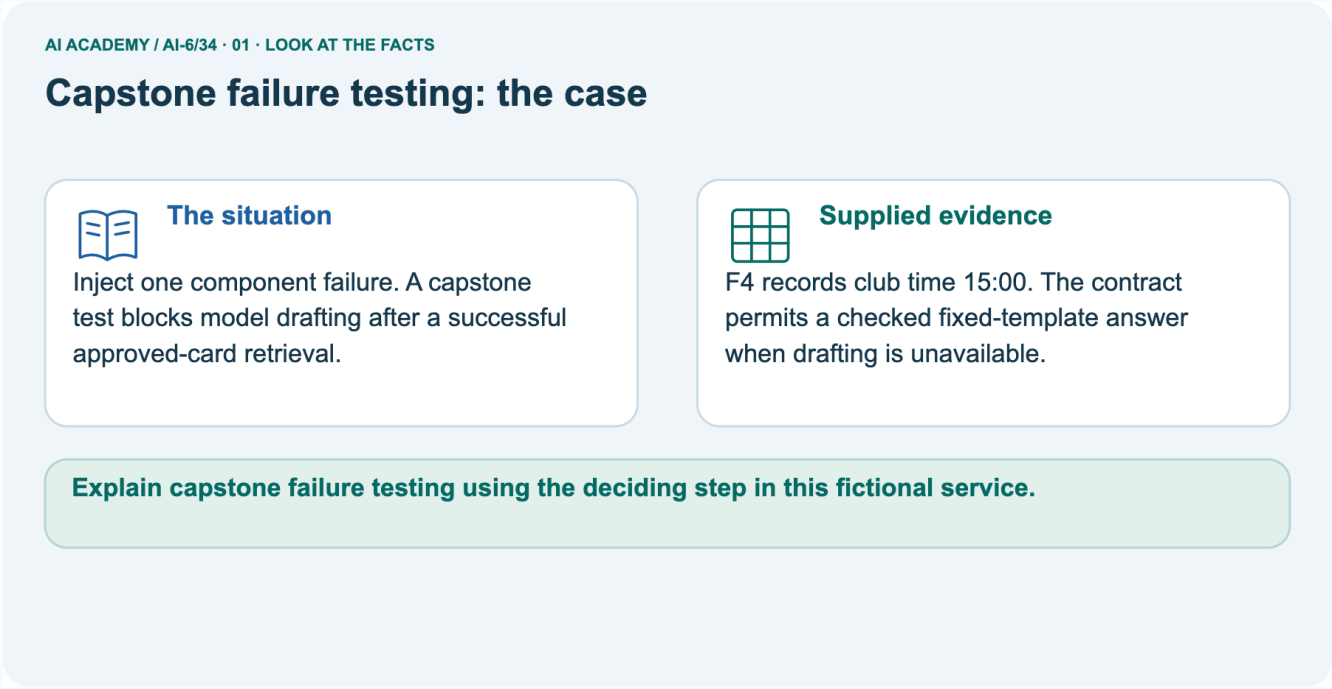
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Capstone failure testing picture from its labels, then record a relevant limit.

01. Capstone failure testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

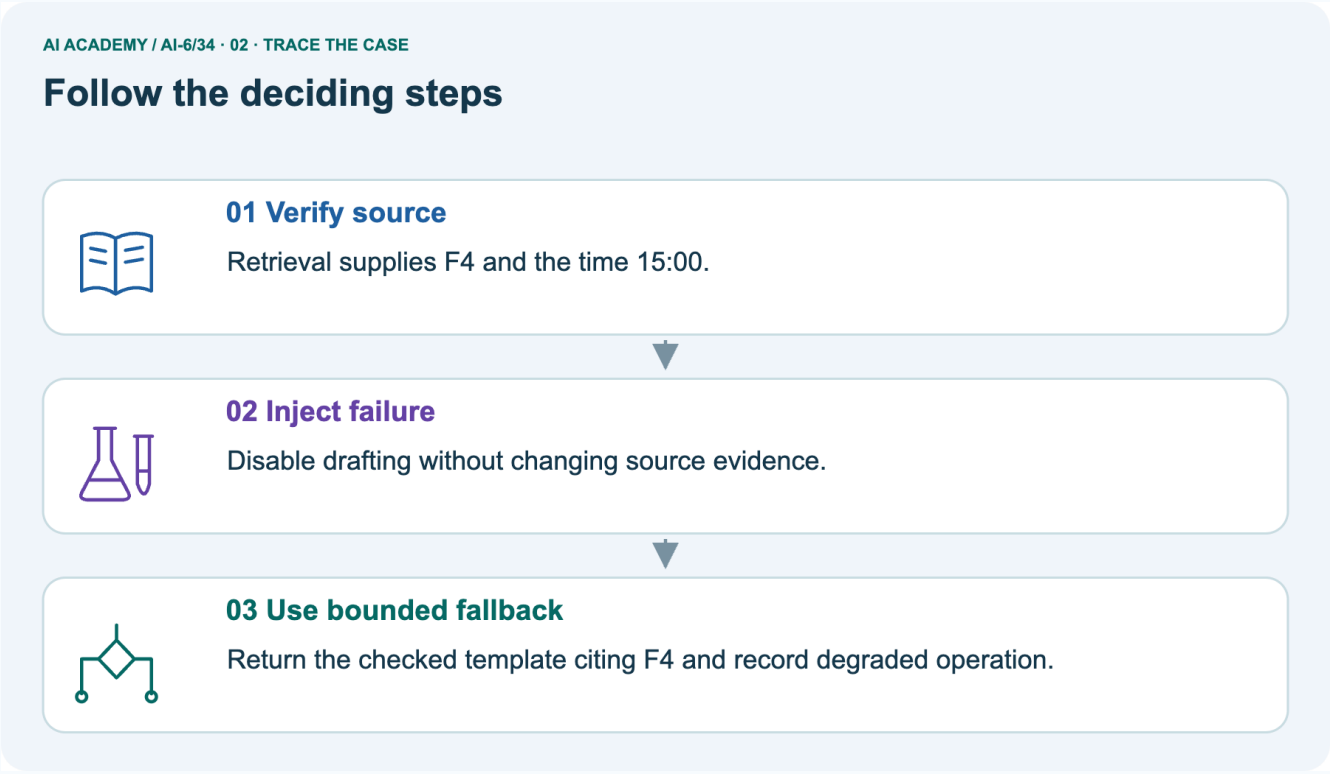
Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval. Supplied facts: F4 records club time 15:00.

The contract permits a checked fixed-template answer when drafting is unavailable.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



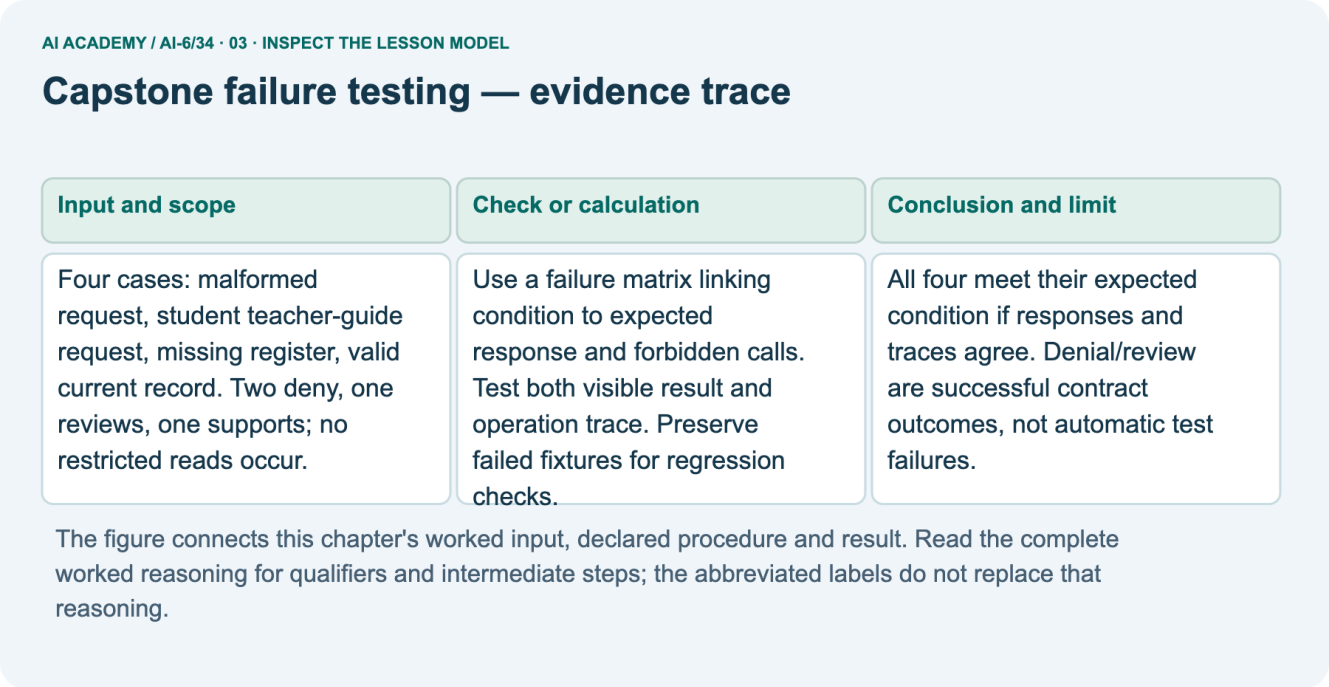
Follow the labelled picture with the matching explanation. Open the vector diagram

Verify source: Retrieval supplies F4 and the time 15:00. → Inject failure: Disable drafting without changing source **evidence**. → Use bounded fallback: Return the checked template citing F4 and record degraded operation.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone failure testing — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone failure testing — **evidence** trace. Four cases: malformed request, student teacher-guide request, missing register, valid current record.

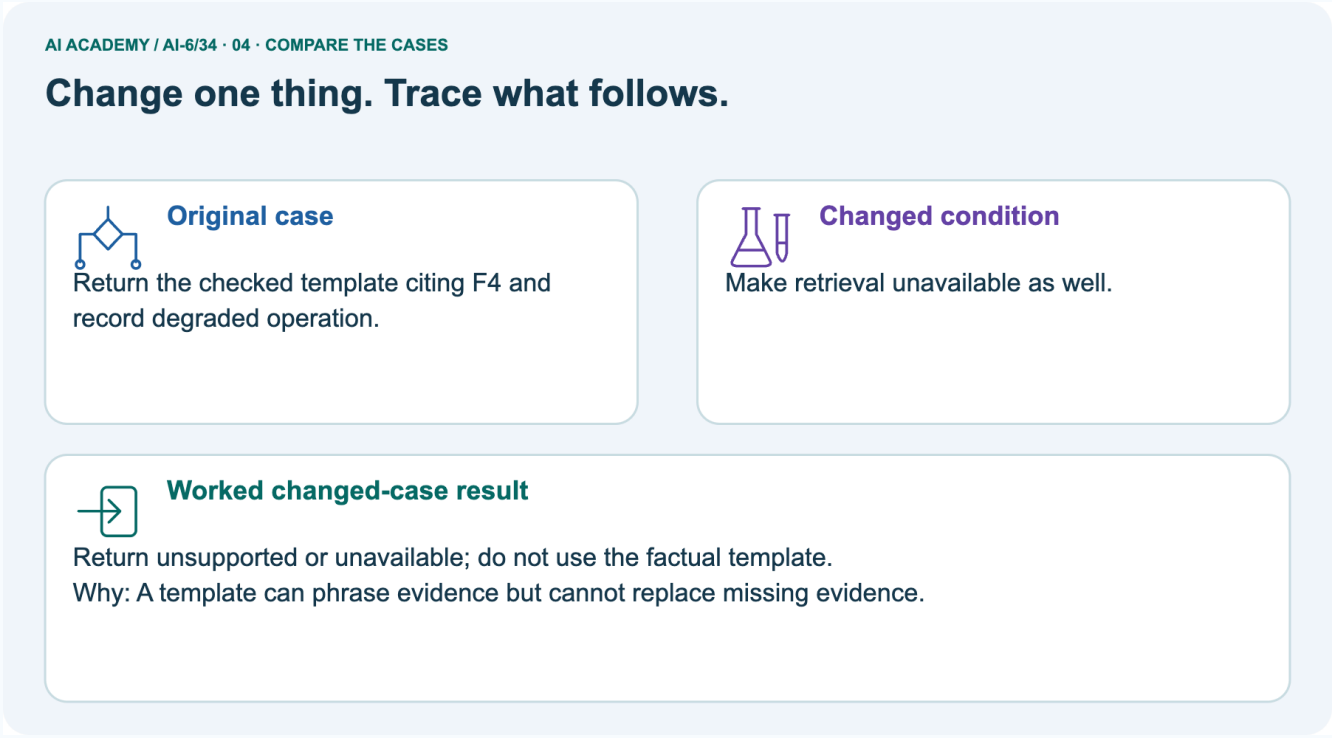
Two deny, one reviews, one supports; no restricted reads occur. | Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace.

Preserve failed fixtures for regression checks. | All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

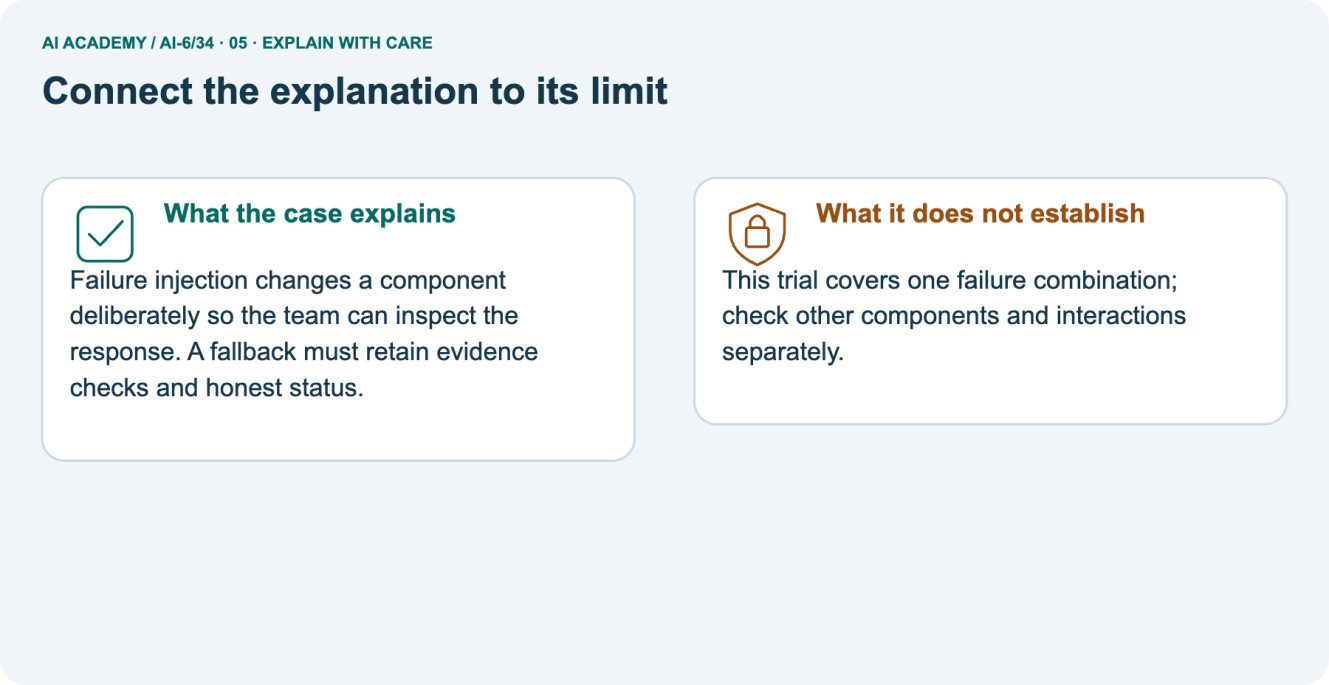
Original case: Return the checked template citing F4 and record degraded operation. Changed condition: Make retrieval unavailable as well. Expected result: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

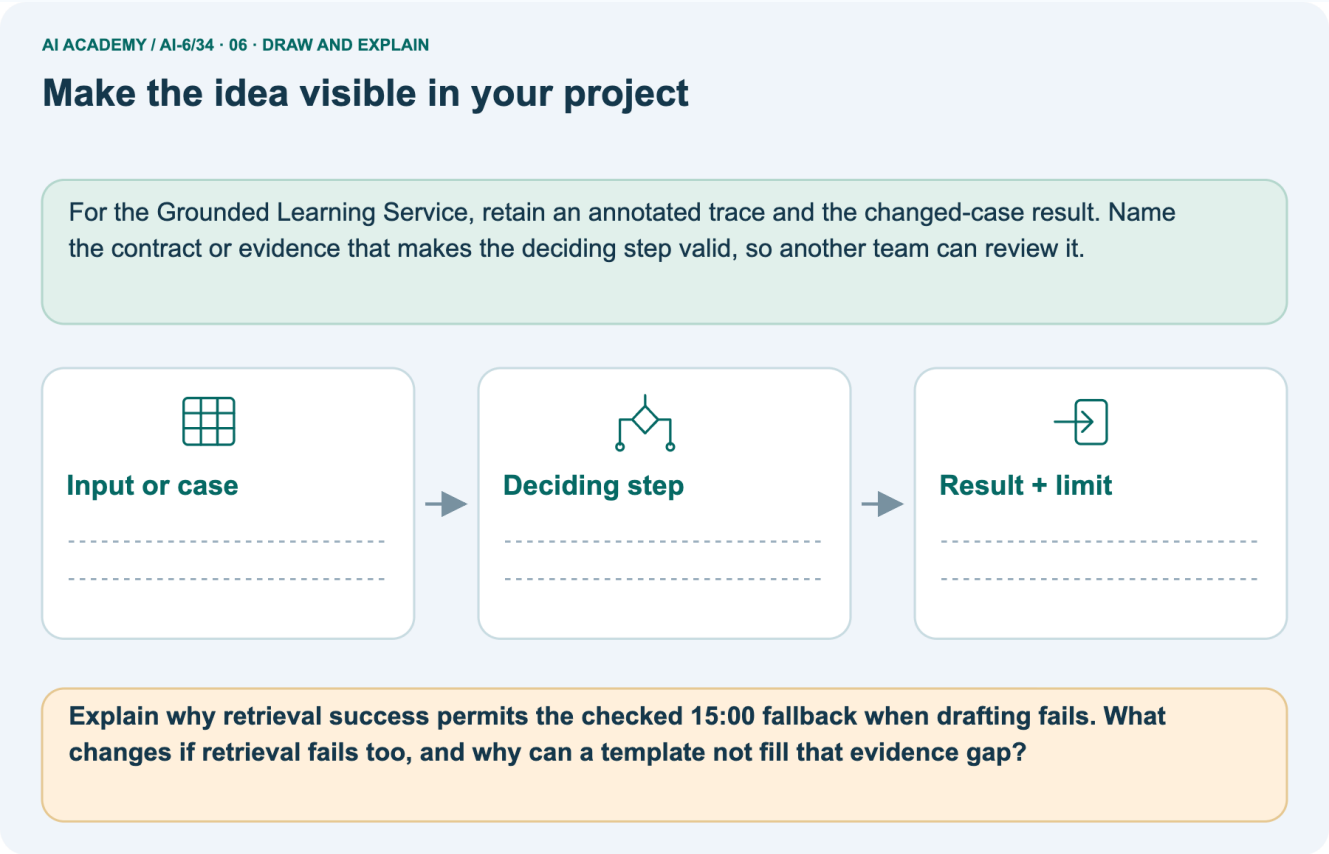
Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Limit: This trial covers one failure combination; check other components and interactions separately.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.